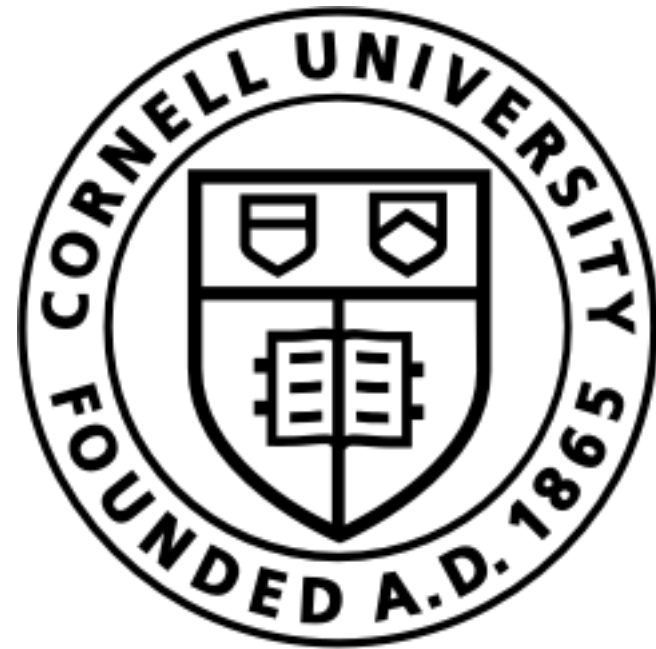




Lecture 4a: Making choices in uncertain situations. Introducing the concept of expected utility

CHEME 5760 Cornell University

Copyright Jeffrey Varner. All Rights Reserved

Objectives. In this lecture, we will:

- ▶ **O1:** Introduce Uncertain, Constrained Resource Allocation Problems and the **expected utility hypothesis**. We'll introduce two forms of this problem for continuous and discrete choices.
- ▶ **O2:** Introduce some basic concepts from probability theory: **discrete random variables, probability spaces, probability mass functions, and discrete expectation and variance**.
- ▶ **O3:** Introduce the Markowitz portfolio optimization problem: Markowitz portfolio optimization is a basic quantitative finance problem: **minimize risk for a specified reward -or- maximize reward for a specified risk**.

L4a long-form notes: <https://bit.ly/3Px1fa1>

Jupyter Notebooks

Repository: <https://github.com/varnerlab/CHEME-5760-Examples-F23.git>

- **Example:** CHEME-5760-L4a-PMFs.ipynb

Concept: Constrained Resource Allocation Problems

An agent has a set of n objects $X = \{x_i\}_{i=1}^n$, a utility function $U : X \rightarrow \mathbb{R}$, and a total of I units of resource to allocate, e.g., money, time, etc. An optimal rational agent maximizes its utility subject to its resource budget:

$$\text{maximize } \mathcal{O} = U(x_1, \dots, x_n) \quad (1)$$

$$\text{subject to } \sum_{i \in 1, \dots, n} c_i x_i = I \quad (2)$$

$$\text{and } x_i \geq 0 \quad i = 1, 2, \dots, n \quad (3)$$

The quantity $c_i \geq 0 \forall i$ denotes the cost of object i , and $x_i \geq 0$ represents the amount of object i purchased or consumed by the agent.

Hmmm. In the maximum utility problem, the **payoff** from a choice *was certain*, i.e., we know it would happen. However, what happens when a payoff isn't certain?

An agent has a set of n objects $X = \{x_i\}_{i=1}^n$, a utility function $U : X \rightarrow \mathbb{R}$, and a total of I units of resource to allocate, e.g., money, time, etc. Each object x_i has a unit cost of $c_i \geq 0$, a probability p_i of occurring. A rational decision-maker maximizes the **expected utility** subject to constraints:

$$\begin{aligned} \text{maximize } \mathbb{E}(x) &= \sum_{k \in 1, \dots, n} p_k U(x_k) \\ \text{subject to } g(x) &\leq 0 \\ \text{and } x_k &\geq 0 \quad k = 1, 2, \dots, n \end{aligned}$$

where $\mathbb{E}(x)$ is the **expectation** computed over the possible alternatives, $g(x)$ is a constraint function, and $x_k \geq 0$ represents the amount of object k purchased or consumed by the agent.

The world is divided into states $\mathcal{S}$, where agents make observations $\mathcal{O}$ and take actions $\mathcal{A}$. Each state $s \in \mathcal{S}$ has a value determined by a utility function $U : \mathcal{S} \rightarrow \mathbb{R}$. Rational agents maximize their expected utility:

$$\begin{aligned} \text{maximize } \mathbb{E}(U(a|o, s)) &= \sum_{s'} P(s'|a, o, s) U(s') \\ \text{subject to } \sum_{s'} P(s'|a, o, s) &= 1 \end{aligned}$$

where $s \in \mathcal{S}$, $a \in \mathcal{A}$, $o \in \mathcal{O}$ and $P(s'|a, o, s)$ is the probability of being in state s' after observing o and taking action a . The principle of maximum expected utility says that a rational agent should choose the action that maximizes expected utility:

$$a^* = \arg \max_a \mathbb{E}(U(a|o, s))$$

O2: Introduce discrete probability spaces, random variables, probability mass functions, and discrete expectation and variance.

Concept: Probability Space

A probability space is completely described by the tuple $(\Omega, \mathcal{F}, P)$:

- Sample space Ω : The set of all possible outcomes from an experiment.
- Event space $\mathcal{F}$: The collection of all possible events. An event E is a subset in Ω that defines an outcome or a combination of outcomes that we are interested in.
- Probability law P : A mapping from an event A to a number $P(A)$ which, measures the size of the event. The probability function P returns a number between 0 and 1.

Concept: Random Variables

A random variable X is a variable whose values are numerical outcomes of a random phenomenon. There are two types of random variables: **discrete** and **continuous** random variables.

- A **discrete** random variable takes on only a countable number of distinct values, i.e., $0, 1, 2, 3, 4, \dots$.
 - If a random variable take only a finite number of distinct values, it is discrete.
 - The probability of observing a specific value is non-negative.
- A **continuous** random variable takes an infinite number of possible values.
 - A continuous random variable is not defined at specific values. Instead, it is defined over an interval of values and is represented by the area under a curve.
 - The probability of observing any single value is equal to zero since the number of values that may be assumed by the random variable is infinite.

Concept: Probability Mass Functions (PMFs)

The probability mass function (PMF) of a discrete random variable X is a function that specifies the probability of obtaining $X = x$, where x is a particular event in the set of possible events we're interested in $\mathcal{F} \subseteq X(\Omega)$:

$$p_X(x) = P(X = x)$$

where $\mathcal{F}$ is the event space, and Ω is the sample space. A probability mass function must satisfy the condition:

$$\sum_{x \in X(\Omega)} p_X(x) = 1$$

Concept: Discrete Expectation

Let X denote a discrete random variable with the probability space $(\Omega, \mathcal{F}, P)$. The expected value of the random variable X is given by:

$$\mathbb{E}[X] = \sum_{x \in \Omega} x p_X(x)$$

where x denotes a value for the discrete random variable X , and $p_X(x)$ denotes the probability of $X = x$. The expectation of a random variable X has the properties:

- $\mathbb{E}(c) = c$ for any constant c
- $\mathbb{E}(cX) = c \times \mathbb{E}(X)$ for any constant c
- $\mathbb{E}(g(X)) = \sum_{x \in X(\Omega)} g(x) p_X(x)$
- $\mathbb{E}(g(X) + h(X)) = \mathbb{E}(g(X)) + \mathbb{E}(h(X))$
- $\mathbb{E}(X + c) = \mathbb{E}(X) + c$ for any constant c

Concept: Variance

Let X denote a discrete random variable with the probability space $(\Omega, \mathcal{F}, P)$. Then, the variance of the random variable X is given by:

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2]$$

where $\mu = \mathbb{E}(X)$ denotes the expected value of the random variable X . The variance of a random variable X has a few interesting (and important) properties:

- $\text{Var}(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2$
- $\text{Var}(cX) = c^2 \text{Var}(X)$ for any constant c
- $\text{Var}(X + c) = \text{Var}(X)$ for any constant c

A Bernoulli random variable X models a binary outcome: either 1 or 0, where 1 occurs with probability p and 0 occurs with probability $1 - p$. The probability mass function (pmf) of the Bernoulli random variable X is:

$$p_X(x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \end{cases} \quad (4)$$

where $0 < p < 1$ is called the Bernoulli parameter. The expectation a Bernoulli random variable equals:

$$\mathbb{E}[X] = p \quad (5)$$

while the variance $\text{Var}(X)$ equals:

$$\text{Var}[X] = p(1 - p) \quad (6)$$

Bernoulli random variables model many binary events: coin flips (H or T), binary bits (1 or 0), true or false, yes or no, present or absent, etc.

Geometric random variables model the number of trials required to obtain the first success in a sequence of independent Bernoulli trials. The probability mass function for a geometric random variable is given by:

$$p_X(k) = (1 - p)^{(k-1)} p \quad k = 1, 2, \dots$$

where p denotes the geometric parameter $0 < p < 1$. The expectation of a geometric random variable X is given by:

$$\mathbb{E}[X] = \frac{1}{p}$$

while the variance $\text{Var}(X)$ is given by:

$$\text{Var}[X] = \frac{1 - p}{p^2}$$

The binomial distribution, the probability of k successes in n independent Bernoulli trials, has the probability mass function:

$$p_X(k) = \binom{n}{k} p^k (1 - p)^{n-k} \quad k = 0, 1, \dots, n$$

where k denotes the number of successes in n independent experiments, the binomial parameter $0 < p < 1$ is the probability of a successful trial and:

$$\binom{n}{k} = \frac{n!}{k! (n - k)!}$$

is the binomial coefficient. The expectation and variance of a binomial random variable is given by:

$$\begin{aligned} \mathbb{E}[X] &= np \\ \text{Var}[X] &= np(1 - p) \end{aligned}$$

Poisson random variables model the number of occurrences of an event in a fixed interval of time or space. The probability mass function for a Poisson random variable is given by:

$$p_X(x) = \frac{\lambda^x}{x!} \exp(-\lambda)$$

where $\lambda > 0$ denotes the Poisson parameter, and $!$ denotes the factorial function. The expectation of a Poisson random variable X is given by:

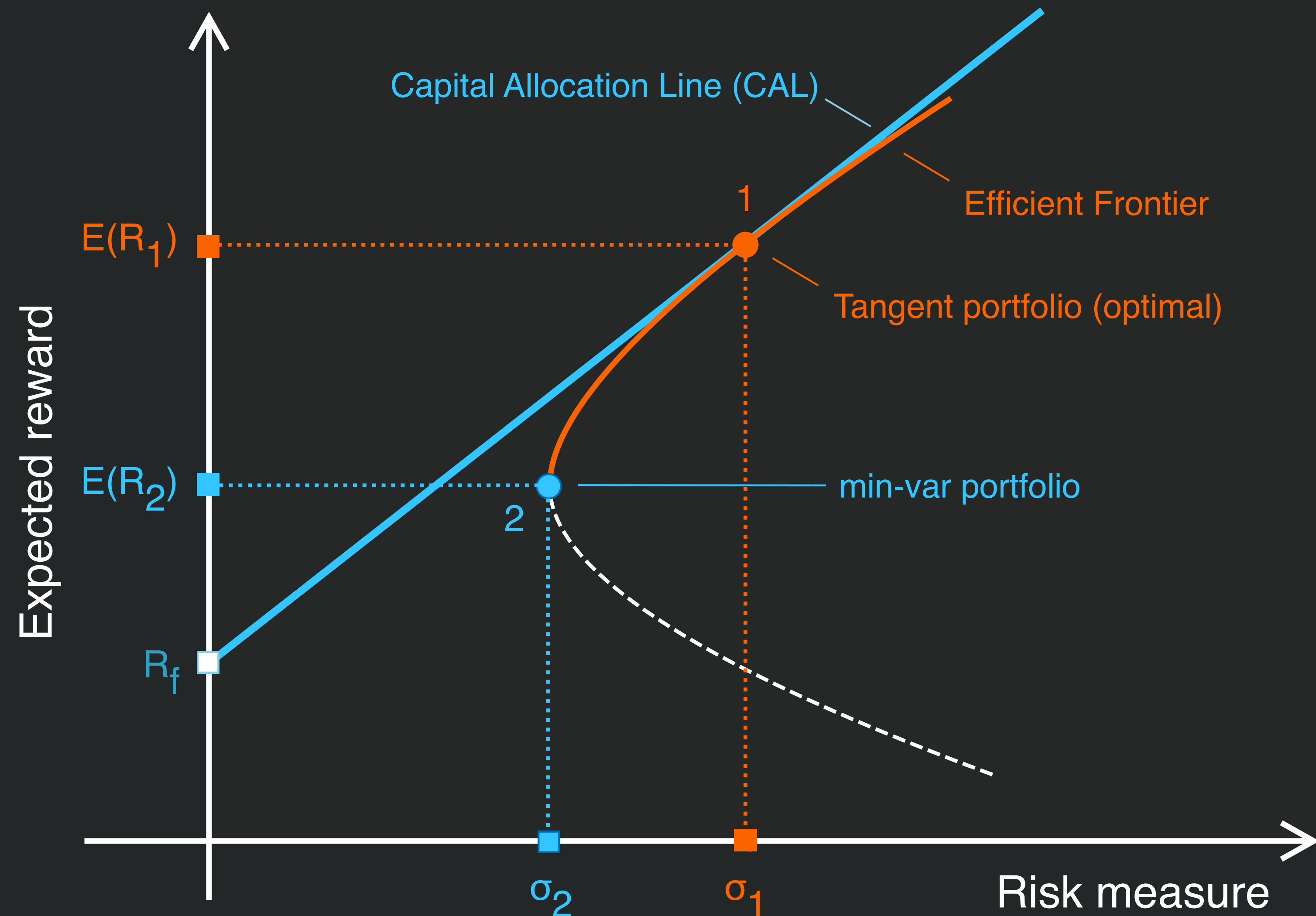
$$\mathbb{E}[X] = \lambda$$

while the variance $\text{Var}(X)$ is given by:

$$\text{Var}[X] = \lambda$$

O3: Markowitz portfolio optimization is a basic quantitative finance application: minimize risk for a specified reward

Optimally balance risk and reward in an asset portfolio. Find the collection of assets that minimizes the risk for a desired level of benefit (return)



Today?

- ▶ We introduced Uncertain, Constrained Resource Allocation Problems and the **expected utility hypothesis**. We introduced two forms of this problem for continuous and discrete choices.
- ▶ We introduced some basic concepts from probability theory: **discrete random variables, probability spaces, probability mass functions, and discrete expectation and variance.**
- ▶ We introduce the Markowitz portfolio optimization problem: Markowitz portfolio optimization is a basic quantitative finance problem: **minimize risk for a specified reward -or- maximize reward for a specified risk.**